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Detection, Localization, and Quantification of a Methane Leak Using Continuous Monitoring and Ensemble Inversion Algorithms.

F. Lascaux and M. Dubois, eLichens, R&D Department, Grenoble, France

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Abstract

A significant part of methane emissions is due to leaks in industrial sites, and European and American regulation are imposing to oil & gas sector to detect and repair gas leaks. Here we present a method based on continuous monitoring of gas concentrations and weather conditions, and ensemble gaussian plume inversion algorithms, with the aim to detect, localize and quantification methane leak in near real-time. We demonstrate that most of the time, the algorithm can localize the leak with a position error inferior to 1 m and quantify the leak rate with relative errors comprised between 16% and 61%.

Introduction

Reduction of methane emissions is a key milestone in the path to reaching climate neutrality in 2050. Regulatory pressures in Europe and North America are accelerating the need for accurate methane leak detection in Oil & Gas operations. In Europe, the 2020 Methane Strategy and the proposed 2021 Regulation on methane emissions [1,2] aim to enforce stricter monitoring and leak detection practices. Similarly, in the U.S., recent EPA rules [3] and provisions under the Inflation Reduction Act [4] seek to significantly reduce methane emissions from oil and gas operations. As a reminder, these are some key figures on methane:

- Methane is the second most important greenhouse gas contributor to climate change [5]
- 60% of the global methane emissions result from human activity [6]
- 1/3 of this comes from the energy sector [7].

Operators are advised that they may use fixed methane detectors as an alternative to conventional detectors (like Optical Gas Imaging - OGI). These continuous monitoring systems must be able to detect and quantify emissions in near-real time. EPA's expected detection sensitivity (2023 update) is 3 kg/h or better, with a low false alarm rate. An effective detection method [8] for methane leak detection and repair (LDAR) should encompass different criteria such as low or zero false positive rate, leak localization, leak quantification, and continuous monitoring. Common methodologies to monitor methane leaks make use of different technologies/approaches:

- Optical gas imaging (OGI) with infrared cameras, widely used for onsite inspections [9]
- Mobile monitoring systems (vehicles, drones equipped with methane sensors [10,11]
- At large scales, satellite-based instruments (like TROPOMI, GHGSat [12])
- Fixed sensors networks allowing for continuous monitoring, combined with inverse modelling or machine learning for gas releases localization and quantification [13,14]. Tests at the Methane Emissions Technology Evaluation Centre (METEC) have demonstrated the capability, under certain conditions, of such systems to identify medium to large leaks within a few meters, with more difficulties for smaller leaks [15].

In this study, we use a network of recently developed ground-based fixed near-dispersion infrared (NDIR) sensors. These autonomous sensors have a sensitivity of ca. 50 ppm that allows for a true continuous monitoring of infrastructures combined with a competitive operational cost – allowing for a dense coverage of the observed site.

Overview of methodology

The approach used in this study is based on continuous monitoring of gas concentrations and weather conditions, and ensemble gaussian plume inversion algorithms. If more than two detectors from the deployed network measure methane concentrations higher than a threshold, their measurements can be used to localize the leak. An ensemble inversion technique of a gaussian plume model is employed. A gaussian plume model is usually adapted for gas clouds at distances higher than 100m from the leak. Here, the gaussian plume model was first fitted to match computational fluid dynamics (CFD) simulated gas clouds in near-field conditions. Once the leak position is known with precision, another ensemble inversion is performed, focusing on retrieving the source leak rate.

In summary, the process of identifying a methane leak rate can be broken in three consecutive steps:

- Detect the leak (alarm)
- Localize the leak (spatial coordinates)
- Quantify the leak (in kg/h)

The whole process can be iterated over depending on the leak duration. The novel steps, with respect to existing solutions, consists in 1) fitting the gaussian plume model parameters to better describe the gas plume near the source (less than 10 m) and 2) using an ensemble inversion technique.

Gaussian plume model fitting

Gas leak modelling is a component of atmospheric dispersion modelling. It involves modelling the dispersion of a gas plume (less dense than air in the case of methane) under different atmospheric conditions in order to predict observable on-site concentrations. Depending on needs and constraints, dispersion models can be divided into two broad categories: gaussian plume dispersion models, or gaussian puff models, and models based on solving partial differential equations of fluid mechanics on finite elements (which we will call CFD for Computational Fluid Dynamics). To retrieve leaks positions and rates, when using continuous monitoring with a network of sensors, one solution consists in inverting such models. The obvious choice is to invert gaussian models, as the equations governing them are well known and easily applicable in short computing time. Their use depends on several underlying assumptions: stationarity, continuous or puff emissions, and the absence of obstacles. Gaussian plume models assume a normal distribution of pollutant concentration in both the horizontal and vertical directions from the source. For a continuous point source, the concentration $C(x,y,z)$ in ppm at a point at coordinates (x,y,z) downwind is given by:

$$C(x,y,z)=\frac{10^6\cdot Q}{\rho_m\cdot 2\pi\cdot\sigma_y\cdot\sigma_z\cdot U}\cdot\exp\left(-\frac{y^2}{2\sigma_y^2}\right)\cdot\left[\exp\left(-\frac{(z-z_n)^2}{2\sigma_z^2}\right)+\exp\left(-\frac{(z+z_n)^2}{2\sigma_z^2}\right)\right] \quad \text{Eq. 1}$$

$$z_n=z_s+\frac{1.6\cdot f\cdot x^{2/3}}{U} \quad \text{Eq. 2}$$

$$f=B^{1/3} \quad \text{Eq. 3}$$

$$B=\frac{g\cdot q_s}{\pi}\left(\frac{1}{\rho_m}-\frac{1}{\rho_a}\right) \quad \text{Eq. 4}$$

Q is the emission rate in kg/s, U is the wind speed in m/s, z_n is the effective stack height adapted from Briggs model in m, B the buoyancy parameter, ρ_m the methane density in kg/m^3, ρ_a the air density in kg/m^3, σ_y, σ_z are the dispersion parameters in the crosswind and vertical directions in m, respectively. The second exponential term accounts for the reflection of the plume from the ground. These models are referenced in environmental studies and regulatory frameworks [16,17]. The dispersion parameters σ_y, σ_z are usually adapted from Pasquill-Gifford-Turner classes [18,19], and are generally valid for distance from the source 100 m to 10 km. They can be described, for example, with a set of parameters (k_0, k_1, k_2, k_3, k_4):

$$\sigma_y=\frac{k_0\cdot x}{\left(1+\frac{x}{k_1}\right)^{k_2}} \quad \text{Eq. 5}$$

$$\sigma_z=\frac{k_3\cdot x}{\left(1+\frac{x}{k_1}\right)^{k_4}} \quad \text{Eq. 6}$$

For every Pasquill atmospheric class, A to F (Table 1), a different set of k_i ($i \in [0,4]$) has been evaluated experimentally (in this study we use the parameters reported in [20]). However, as already mentioned, they are valid for distances to the source from 100 m to 10 km. As the methane sensors used in this study can measure concentrations close to the source (< 10 m), we need to refine the distribution parameters for these distances where the plume behavior is dominated by source geometry and initial momentum. Our hypothesis is that it is possible to find different sets of parameters k_i that will describe the plume close to the source, for different wind conditions.

Table 1—Description of the different Pasquill stability classes

Stability Class	Description	Typical Meteorological Conditions
A	Very unstable	Strong solar radiation, light winds
B	Unstable	Moderate solar radiation, light to moderate winds
C	Slightly unstable	Weak solar radiation or overcast, moderate winds
D	Neutral	Overcast conditions, moderate to strong winds
E	Slightly stable	Clear night, light winds
F	Stable	Clear night, very light winds

Detecting the leak

In controlled releases (CR), a leak is detected by the network of sensors when at least one sensor measures concentrations higher than a given threshold for a minimum amount of time (e.g. 10 ppm over a 15-min period). On industrial sites with non-negligible background values, it would be of interest to evaluate those background concentrations (not done in this study).

Localizing the leak

If two sensors at least detect an occurring leak, we can launch the localization algorithm, using wind measurements and concentrations readings (over a 15-min average). In this inversion process, we use the previously fitted gaussian model. The localization algorithm is an optimization algorithm for minimizing scalar functions of one or more variables. The goal in this first step is to find the coordinates of the leak that minimize the mean square error (MSE) between measurements and a gaussian plume model previously fitted for the current wind conditions. We choose whether the Nelder-Mead or the Powell methods from the Python package SciPy [21], as they don't require derivatives and work well for smooth functions. However, they are sensitive to the initial conditions and convergence to a global minimum is not guaranteed. To enhance the probability of finding the good minimum (the most adequate leak coordinates), we vary the initial conditions and use different initial parameters for the methods, in a so-called ensemble inversion technique. An analysis of the results will then allow us to identify the best candidate for the leak coordinates. All the computed candidates are united in clusters, and the final candidate is the center of the biggest cluster. An example of output is visible Fig. 1.

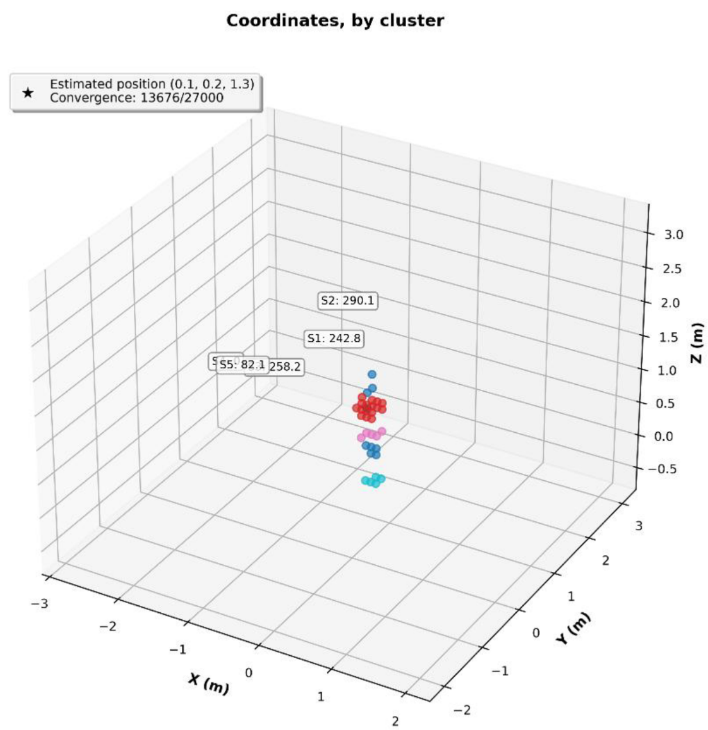

Figure 1—Example of all the computed coordinates of a methane leak using the ensemble inversion technique. All the coordinates are grouped inside clusters, and the best candidate for the leak is the center of the biggest cluster (here in red). The number inside boxes are sensors readings in ppm, and the real leak was at the coordinates (0,0,1.4). The best candidate in this case was found at the coordinates (0.1,0.2,1.3).

Quantifying the leak

Once the leak coordinates are identified, we perform a second minimization to estimate the leak rate. As with the localization process, we use an ensemble inversion technique, varying the initial conditions (particularly around the previously estimated coordinates). From the resulting ensemble of leak rates (sequenced with histograms), we fit a distribution to the outputs. The leak rate is then estimated as the mode (peak) of this distribution. The confidence interval is derived from the Full Width at Half Maximum (FWHM), indicating the spread around the peak.

Results

Gaussian plume model fitting

We fit the dispersion parameters k (Eq. 5 and Eq. 6) to CFD simulations under different wind velocities, using a minimization technique. CFD simulations used as references are run with the OpenFoam software[1], with adapted solvers for gas leak dispersion. Results for wind speeds of 1 m/s, 2 m/s and 3 m/s are reported in Table 2.

Table 2—List of parameters k_i defining the Gaussian models' dispersion parameters (see Eq. 2 and Eq. 3), under different wind velocities, close to the point source. As a comparison, we also reported the values for Pasquill atmospheric class C.

WIND SPEED	k_1	k_2	k_3	k_4	k_5
1 m/s	16.459	0.039	1.036	0.178	-0.054
2 m/s	0.706	15.072	-0.134	0.065	-2.164
3 m/s	0.291	2.580	0.103	0.284	0.759
Class C	0.134	283	0.134	0.0722	0.102

Ensemble inversion technique: localization and quantification

We have conducted some controlled releases experiments on a biomethane site. In total, 7 releases of around 15-min each, with different leak rates between 0.03 kg/h and 1 kg/h (small to medium leak rates), and different sensors configurations, have been set up. The list of the leaks is reported Table 3.

Table 3—List of configurations of all controlled releases experiments conducted in this study. Leaks 5 and 5b are the same cases, except that the closest sensor (20 cm to the source) was removed before the inversion algorithms in case 5b.

Leak Number	Leak Rate (kg/h)	Number of deployed sensors	Distances sensors-leak
1	0.03	3	0.2 – 2 m
2	0.07	3	0.2 – 2 m
3	0.20	5	0.2 – 2.2 m
4	0.50	5	0.2 – 2.2 m
5	1.00	5	0.2 – 2.5 m
5b	1.00	4	1.0 – 2.5 m
6	0.63	5	1.5 – 3 m
7	0.31	5	1.5 – 3 m

To consider the sensors' errors, we add an error of +/- 50 ppm to each sensor reading (in link with the current sensor's sensitivity of 50 ppm), and list all the associated combinations (for n sensors, the total number of combinations is $N = 3_n$). The inversion algorithm is performed for each of these combinations, or at least a randomized sub-sample ($P<N$, to speed up the computations). The results in terms of detection, localization and quantification are reported Table 4 (P=15), Table 5 (P=5) and Table 6 (P=1). When detected, all leaks were successfully localized with an error of less than 1 m, and the relative errors of the quantification (when the model has converged) are all around 50% for P=15 (from 16% to 61%). For P=5, localization errors are less than 1.7 m, and the relative errors of the quantification reach 104% (leak #6). An example of the representation of the outputs for one inversion, for leak #5, is visible Figure 2. Having a higher number of sensor combinations enhances the probability of the ensemble algorithm to find a solution, however a reduced number of sensors combinations will speed up the computing time. It is worth noting that not

simulating sensors' errors can lead to bad estimates of the location, leading to failure in quantifying the leak rate (see for example leak #7 in Table 5).

Table 4—Results of the detection, localization and quantification of methane leaks for different controlled releases. Leaks #1 and #2 were not detected by the sensors (too small a leak rate, <0.1 kg/h). Leaks #3 and #4 were detected and localized with an accuracy < 0.5 m, but the inversion for the quantification did not converge towards a solution. In the cases presented here, a random sub-sample of P=15 combinations of sensors (including an error of +/− 50 ppm) was used in the ensemble inverse modelling.

Leak Number	Detected	Seen by	Position Error	Real Leak rate (kg/h)	Modelled Leak Rate (kg/h); P=15	FWHM (kg/h); P=15
3	Yes	1 sensor	0.5 m	0.20	N/D	N/D
4	Yes	2 sensors	0.2 m	0.50	N/D	N/D
5	Yes	5 sensors	0.2 m	1.00	1.47	[0.38,3.15]
5b	Yes	4 sensors	0.7 m	1.00	0.84	[0.30,2.45]
6	Yes	3 sensors	1.0 m	0.63	1.03	[0.56,1.52]
7	Yes	2 sensors	1.0 m	0.31	0.50	[0.25,1.00]

Table 5—Same as Table, but with a random sub-sample of P=5 sensors combinations. Note: the high error for leak #3 position comes from the vertical component error.

Leak Number	Detected	Seen by	Position Error	Real Leak rate (kg/h)	Modelled Leak Rate (kg/h); P=5	FWHM (kg/h); P=5
3	Yes	1 sensor	5.4 m	0.20	N/D	N/D
4	Yes	2 sensors	0.3 m	0.50	N/D	N/D
5	Yes	5 sensors	0.1 m	1.00	1.36	[0.30,3.10]
5b	Yes	4 sensors	0.6 m	1.00	0.88	[0.21,2.34]
6	Yes	3 sensors	0.8 m	0.63	1.29	[0.86,1.93]
7	Yes	2 sensors	1.7 m	0.31	0.38	[0.17,0.98]

Table 6—Same as Table and Table but without considering any error on sensors' readings (i.e. P=1).

Leak Number	Detected	Seen by	Position Error	Real Leak rate (kg/h)	Modelled Leak Rate (kg/h); P=1 (no error)	FWHM (kg/h); P=1
3	Yes	1 sensor	4.7 m	0.20	N/D	N/D
4	Yes	2 sensors	0.3 m	0.50	N/D	N/D
5	Yes	5 sensors	0.2 m	1.00	1.48	[0.37,3.20]
5b	Yes	4 sensors	1.0 m	1.00	1.92	[0.83,3.75]
6	Yes	3 sensors	0.5 m	0.63	0.98	[0.01,2.02]
7	Yes	2 sensors	7.3 m	0.31	N/D	N/D

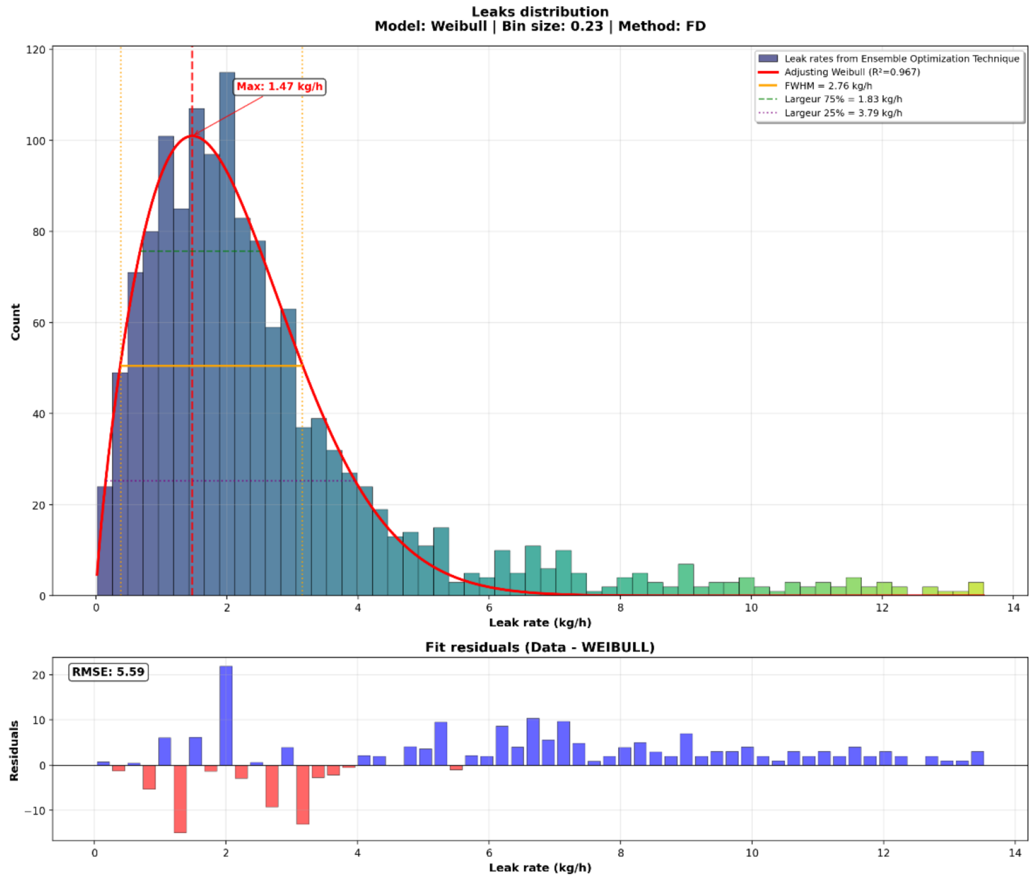

Figure 2—Histogram of all the outputs from the inversion ensemble modelling of the leak rate for leak #5 with P=15. The best distribution fitting the data (in terms of minimum RMSE) is the Weibull distribution. The x-value at peak is 1.47 kg/h, which is the best candidate for the modelled leak rate (the one with the highest probability).

The convergence of the algorithm towards a reliable solution depends on several factors:

- First, a good estimation of the position of the leak is critical for the convergence of the quantification algorithm
- A good estimation of the position is highly correlated with the number of activated sensors (at least 2)
- Accounting for the sensors' errors enhances the probability of convergence of the quantification algorithm

Conclusion

In this paper, we have demonstrated the reliability of our solution for detecting and localizing methane leaks, as well as quantifying the associated leak rates with strong accuracy, by combining continuous monitoring and an ensemble gaussian plume model inversion technique. An analysis of these first tests shows that:

- Accounting for sensors' errors is critical for the convergence of the ensemble algorithm
- For the detected leaks, the localization error remained below 1 m, and the quantification relative error was around 50% for P=15. Reducing the sample size P of sensor combinations may result in less robust outcomes, with higher errors both in localization and quantification. The optimal number of combinations may depend on the number of sensors detecting the leak, and further investigation is required to define it.

Having established the reliability of the solution, the next steps will concentrate on extending validation and supporting industrial integration:

- Evaluate performance under varying background methane concentrations to confirm stability across diverse operational contexts
- Undertake a broader field campaign under evolving environmental conditions (including blind tests) to showcase the robustness of the solution, strengthen confidence, and tailor the system to different industrial use cases for large-scale deployment.

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